Unit - V

Statistical Learning and Applications

1. Learning objectives

After studying this topic, students should be able to:

- Explain statistical learning and its relationship with artificial intelligence, machine learning, and data science.
- Distinguish supervised and unsupervised learning.
- Formulate regression problems mathematically.
- Explain linear and logistic regression.
- Describe bias, variance, overfitting, and underfitting.
- Apply the basic idea of training, validation, testing, and cross-validation.
- Select suitable evaluation metrics for real-world applications.
- Identify practical uses of statistical learning in engineering and industry.

2. What is statistical learning?

Statistical learning is a collection of methods used to understand relationships between variables and to make predictions or decisions from data.

Suppose we observe input variables: $X = (X_1,\ X_2, X_3, \ldots\ldots\ X_p)$ and an output variable Y.

Statistical learning attempts to estimate an unknown relationship: $Y = f(X) + \varepsilon$

where:

- f(X) is the systematic relationship between inputs and output.
- ϵ is random noise or an error term.
- X represents features or predictor variables.
- Y represents the target or response variable.

The objective is usually to estimate f(x) using observed data and then use the estimated function $\hat{f}$ to predict new observations: $\hat{Y} = \hat{f}(X)$

Example: A college may want to predict student placement salary using:

- CGPA.
- Internship experience.
- Programming skill score.
- Communication score.
- Number of projects.
- Department.

The features are X, and salary is Y.
A statistical learning model estimates the relationship between these variables.

3. Statistical learning and related fields

Field	Main goal	Typical question
Statistics	Understand uncertainty and relationships in data	Which variables are related to income?
Machine learning	Build systems that improve predictions from data	Can we predict whether a customer will leave?
Artificial intelligence	Build systems that perform tasks requiring intelligent behaviour	Can a system diagnose a disease or understand speech?
Data science	Use data, computing, statistics, and domain knowledge to solve problems	How can a company improve sales using customer data?
Deep learning	Learn complex representations using multilayer neural networks	Can an image system identify traffic signs?

These fields overlap significantly. Statistical learning provides many of the principles used in classical machine learning, including regression, classification, regularization, and model validation.

4. Types of statistical learning

Statistical learning can broadly be divided into two categories.

(i) Supervised learning
(ii) Unsupervised learning

4.1 Supervised learning

In supervised learning, the training data contains both input features and known output values.

$(X_1,\ Y_1),\ (X_2,\ Y_2),\ (X_3,\ Y_3), \dots\dots, (X_n,\ Y_n)$

The model learns a mapping from X to Y.

Main supervised-learning tasks

1) **Regression:** Predict a numerical value.
2) **Classification:** Predict a category or class.

Examples:

- Predict house price: regression.
- Predict whether a transaction is fraudulent: classification.
- Predict temperature tomorrow: regression.
- Predict whether an email is spam: classification.

4.2 Unsupervised learning

In unsupervised learning, the output labels are not provided. The model attempts to find structure in the data.

Examples:

- Customer segmentation.
- Grouping similar documents.
- Detecting unusual network activity.
- Finding common patterns in shopping behaviour.

Common techniques include clustering, dimensionality reduction, and association-rule mining.

5. Regression

Regression is a supervised learning technique used to predict continuous numerical values by learning relationships between input variables (features) and an output variable (target). It helps understand how changes in one or more factors influence a measurable outcome and is widely used in forecasting, risk analysis, decision-making and trend estimation.

Linear Regression is a fundamental supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It predicts continuous values by fitting a straight line that best represents the data.

- It assumes that there is a linear relationship between the input and output
- Uses a best-fit line to make predictions
- Commonly used in forecasting, trend analysis, and predictive modelling

Other examples include:

- Predicting electricity consumption.
- Estimating automobile resale value.
- Forecasting product demand.
- Forecasting Sales for a Business
- Predicting delivery time.

The general regression model is: $Y = f(X) + \varepsilon$

The model estimates $\hat{f}$, and the prediction is: $\hat{Y} = \hat{f}\ (\mathrm{X})$

5.1 Linear regression

Linear Regression is a fundamental supervised learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It predicts continuous values by fitting a straight line that best represents the data.

- It assumes that there is a linear relationship between the input and output
- Uses a best-fit line to make predictions
- Commonly used in forecasting, trend analysis, and predictive modelling

5.1.1 Simple linear regression

Simple linear regression models the relationship between a dependent variable and a single independent variable.

Simple Linear Regression aims to describe how one variable i.e., the dependent variable changes in relation with reference to the independent variable.

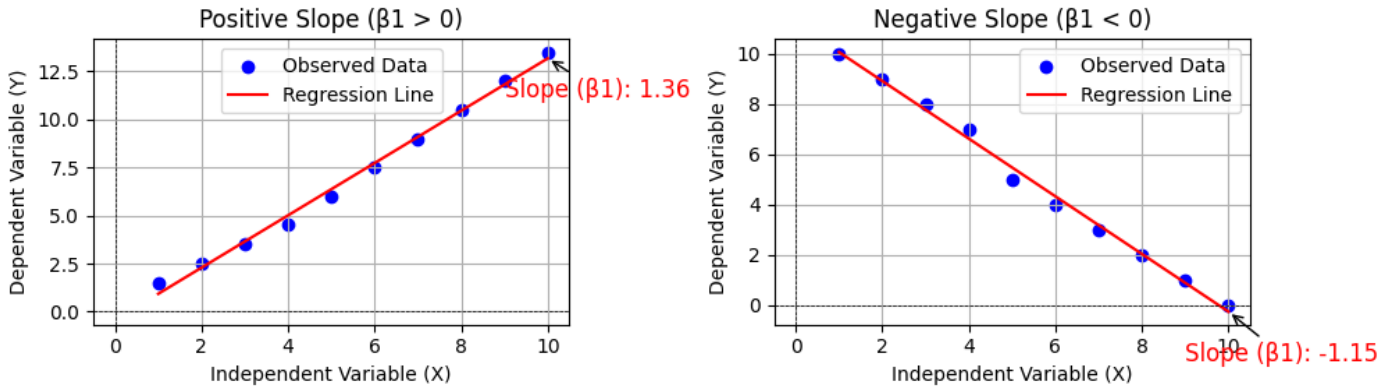
For example, consider a scenario where a company wants to predict sales based on advertising expenditure. By using simple linear regression, the company can determine if an increase in advertising leads to higher sales or not.

Simple linear regression uses one input variable: $Y = \beta_0 + \beta_1 X + \varepsilon$

where:

- β0 is the intercept.
- β1 is the slope.
- X is the input feature.
- Y is the target.
- ϵ is the random error.

The estimated prediction equation is: $\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1}\ X$



Fitting of a linear equation Y = a + b X to the given data:

Let (*Xi*, *Yi*), *i* = 1, 2, ……, *n* be *n* sets of observations of related data and Y = a + b X be the straight line to be fitted.

The residual at *X* = *Xi* is E*i* = Y*i* – *f*(*Xi*) = *Yi* – *a* – *b Xi*

Introduce a new quantity U such that

$$U = \sum E_i^2 = \sum (Y_i - a - b\,X_i)^2$$

By the principle of Least squares, U is minimum.

$\frac{\partial U}{\partial a} = 0$, and $\frac{\partial U}{\partial b} = 0$

$\frac{\partial U}{\partial a} = 0$ $\quad 2 \sum (Y_i - a - b\,X_i)(-1) = 0$ $\quad \sum Y_i = n\,a + b \sum X_i$ ---------- (1)

$\frac{\partial U}{\partial b} = 0$ $\quad 2 \sum (Y_i - a - b\,X_i)(-Xi) = 0$ $\quad \sum X_i Y_i = a\,X_i + b \sum X_i^2$ -------- (2)

Since $\sum X_i$, $\sum Y_i$, $\sum X_i Y_i$, $\sum x_i^2$ are substitute in the equations (1) and (2) then solve for the values of a and *b*.

Then we get the fitted regression line equation $\hat{Y}$ = *a* + *bx*.

Example 1: The following data represents the number of hours, 12 different students watched television during the weekend and the scores of each student who took a test the following Monday.

a) Find the equation of the regression line.
b) Use the equation to find the expected test score for a student who watches 9 hours of TV.

Hours, X	0	1	2	3	3	5	5	5	6	7	7	10
Test Score, Y	96	85	82	74	95	68	76	84	58	65	75	50

Solution:

Hours, X	0	1	2	3	3	5	5	5	6	7	7	10	∑X=54
Test Score, Y	96	85	82	74	95	68	76	84	58	65	75	50	∑Y=908
XY	0	85	164	222	285	340	380	420	348	455	525	500	∑XY=3724
X^2	0	1	4	9	9	25	25	25	36	49	49	100	$\sum X^2$= 332

The Normal equations of Y = a + b X are $\sum Y_i = n\,a + b \sum X_i$ ---------- (1)

$$\sum X_i Y_i = a\,X_i + b \sum X_i^2 \text{ ------- (2)}$$

Substitute $\sum X_i$, $\sum Y_i$, $\sum X_i Y_i$, $\sum x_i^2$ values in the normal equations, then we get

908 = 12 a + 54 b -------- (3)

3724 = 54 a + 332 b ------- (4)

Solve eq (3) & eq (4) for the values of a & b, then we get a = 93.97, and b = - 4.067

Then the predicted linear regression line is $\hat{Y} = 93.97 - 4.067\,(X)$

Using the equation $\hat{Y} = 93.97 - 4.067\,(X)$, we can predict the test score for a student who watches 9 hours of TV. Is $\hat{Y} = 93.97 - 4.067\,(9) = 57.34$

A student who watches 9 hours of TV over the weekend can expect to receive about 57 marks (Test Score) on Monday's test.

5.1.2 Multiple linear regression

When several features are used, the model becomes:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

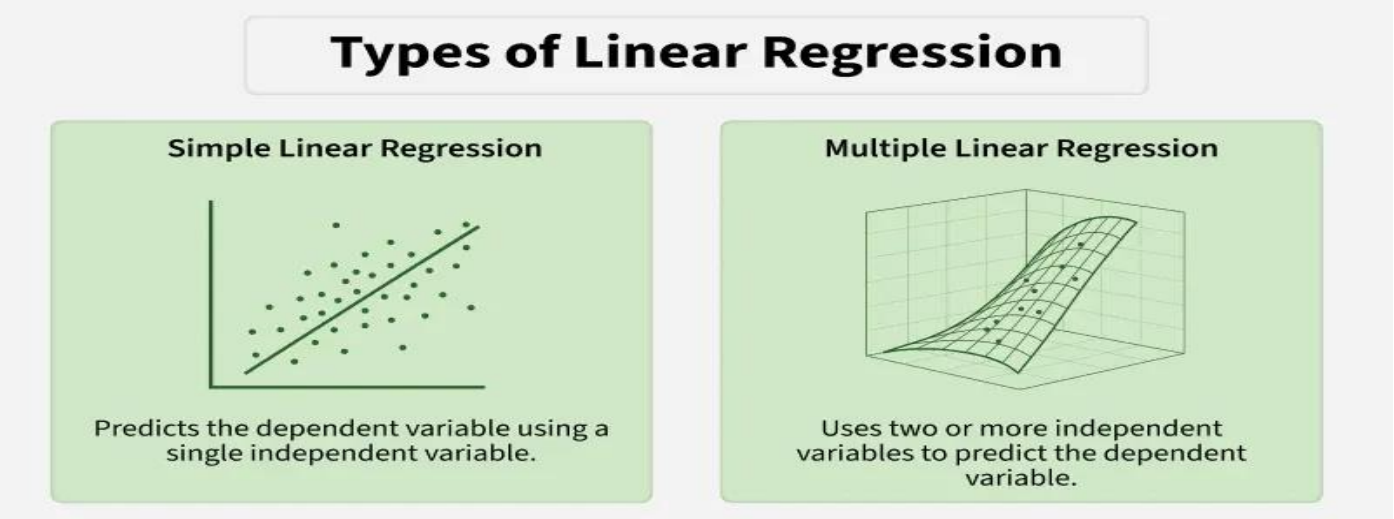
The prediction equation is:

$$\hat{Y} = \widehat{\beta_0} + \widehat{\beta_1}\ X_1 + \widehat{\beta_2}\ X_2 + \cdots + \widehat{\beta_p}\ X_p$$

Example: house-price prediction

Price = β0 + β1 (area) + β2 (bedrooms) + β3 (age) + β4 (distance from city)

A positive coefficient indicates that an increase in the corresponding feature tends to increase the predicted target, while a negative coefficient indicates a decreasing relationship, assuming other variables remain fixed.



5.1.3 Estimating coefficients using least squares

The most common method for fitting linear regression is **ordinary least squares**.

For observation iii, the residual is:

$$e_i = Y_i - \hat{Y}_i$$

where e_i is the prediction error or residual.

OLS chooses the coefficients that minimize:

$$SSE = \sum_{i=1}^{n} \left(Y_i - \hat{Y}_i\right)^2$$

A large error is penalized more heavily because errors are squared.

Why minimize squared error?

- It produces a mathematically convenient optimization problem.
- It gives greater importance to large errors.
- It has a clear geometric interpretation.
- It works well when errors are approximately normally distributed.

Limitations of linear regression

Linear regression may perform poorly when:

- The true relationship is strongly nonlinear.
- Important features are missing.
- Features are highly correlated.
- Extreme outliers are present.
- The data contains changing patterns over time.
- The training data is not representative of future data.

5.1.4 Regression evaluation metrics

Mean squared error

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)^2$$

MSE gives more weight to large errors.

Root mean squared error:

$$RMSE = \sqrt{MSE}$$

RMSE has the same units as the target variable.

For example, if the target is salary in thousands of rupees, RMSE is also measured in thousands of rupees.

Mean absolute error:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|Y_i - \hat{Y}_i\right|$$

MAE is easier to interpret and is less affected by extreme errors than MSE.

Coefficient of determination:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(Y_i - \hat{Y}_i\right)}{\sum_{i=1}^{n}\left(Y_i - \bar{Y}_i\right)}$$

R^2 measures the proportion of variation in the target explained by the model.

For example, R^2 = 0.80 means that the model explains 80% of the variation in the observed target under the assumptions of the evaluation setting.

A high R^2 does not necessarily imply that:

- The model is causally correct.
- The model will perform well on new data.
- The features are unbiased.
- The model is appropriate for decision-making.

6. Logistic regression

Logistic regression is a widely used supervised learning algorithm used for classification tasks. It helps model the relationship between input features and a categorical outcome by estimating class probabilities, making it simple, efficient and easy to interpret.

- Used for binary and multiclass classification
- Predicts probabilities using the logistic (sigmoid) function

It is useful when the target has two possible outcomes, such as:

- Pass or fail.
- Spam or not spam.
- Fraud or legitimate.
- Disease or no disease.
- Customer churn or retention.

Let: $Y \in \{0, 1\}$ Logistic regression estimates the probability that Y = 1.

6.1 The logistic function

First, logistic regression calculates a linear score:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

It then converts this score into a probability using the **Sigmoid function:**

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-z}}$$

The output is always between 0 and 1.

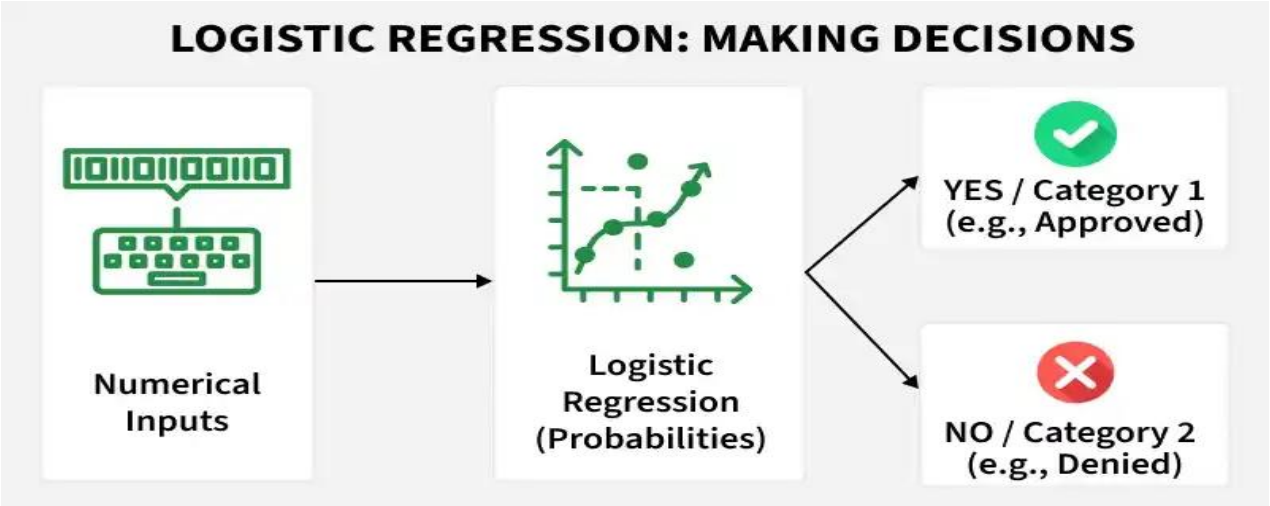
Sigmoid behaviour

- If z is very large, the probability approaches 1.
- If z is very negative, the probability approaches 0.
- If z=0, the probability is 0.5.

A classification threshold is then applied:

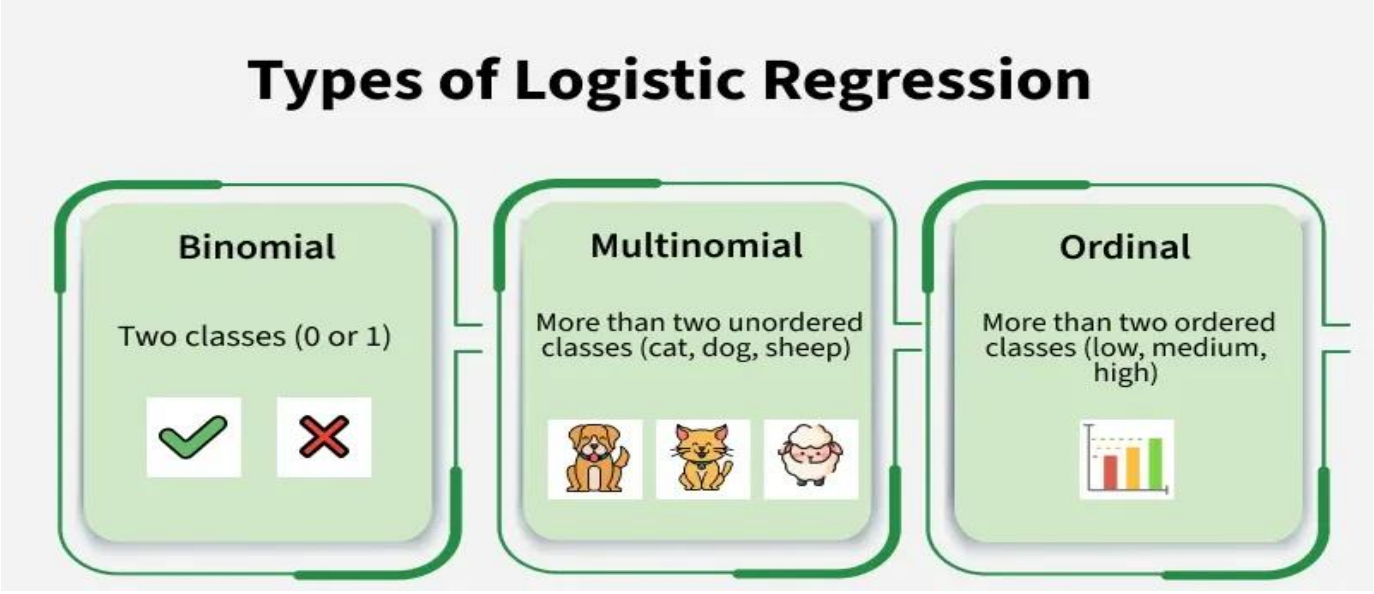
$$\hat{Y} = \begin{cases} 1, & if\ P(Y = 1 \mid X) \geq 0.5 \\ 0, & Otherwise \end{cases}$$

The threshold does not always need to be 0.5. In medical screening, for example, a lower threshold may be used to reduce missed positive cases.



Types

1. **Binomial Logistic Regression:** Used when the dependent variable has only two possible categories, such as Yes/No, Pass/Fail, or 0/1. It is the most common type and is used for binary classification tasks.
2. **Multinomial Logistic Regression:** Used when the dependent variable has three or more unordered categories. For example, classifying animals as cat, dog, or sheep. It extends logistic regression to handle multiple classes.
3. **Ordinal Logistic Regression:** Used when the dependent variable has three or more categories with a natural order, such as Low, Medium, and High. It considers the ranking of categories during prediction.



Understanding Sigmoid Function

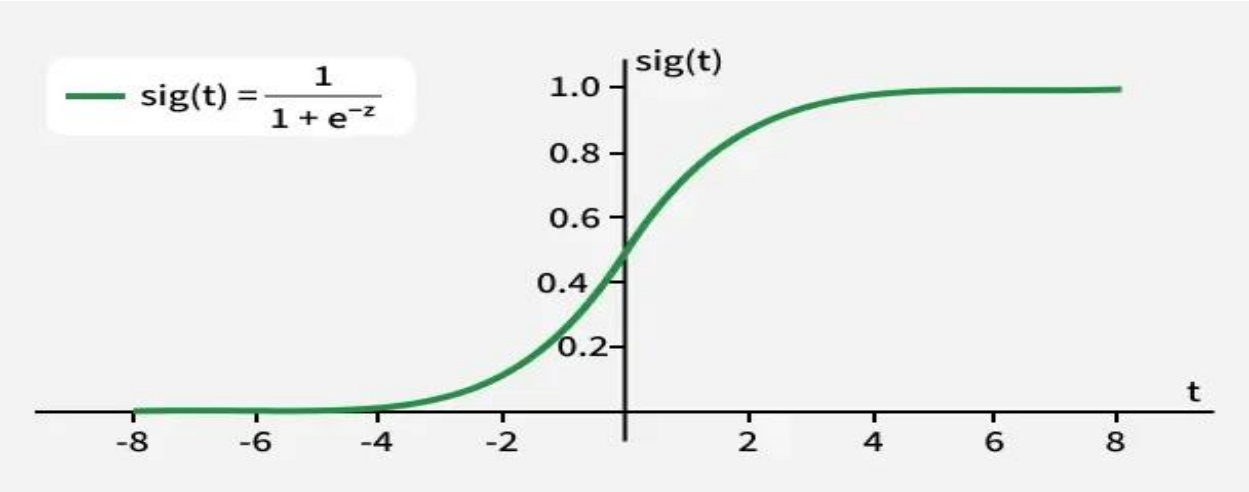
1. The sigmoid function is a key component of Logistic Regression that converts the model's raw output into a probability value between 0 and 1.

2. This function takes any real number and maps it into the range 0 to 1 forming an "S" shaped curve called the sigmoid curve or logistic curve. Because probabilities must lie between 0 and 1, the sigmoid function is perfect for this purpose.

3. In logistic regression, we use a threshold value usually 0.5 to decide the class label.

- If the sigmoid output is same or above the threshold, the input is classified as Class 1.
- If it is below the threshold, the input is classified as Class 0.

  This approach helps to transform continuous input values into meaningful class predictions.



6.2 Confusion Matrix: A **confusion matrix** is a table used to evaluate a **classification model**, such as logistic regression.

It compares: **Actual Class**

With

Predicted class

For example.

- Pass = 1
- Fail = 0
- Suppose our logistic regression model makes the following predictions:

Student	Actual	Predicted
S1	Pass	Pass
S2	Pass	Pass
S3	Pass	Pass
S4	Pass	Fail
S5	Fail	Fail

The basic confusion matrix is:

Actual \ Predicted	Pass (1)	Fail (0)
Pass (1)	True Positive (TP)	False Negative (FN)
Fail (0)	False Positive (FP)	True Negative (TN)

6.2.1 Evaluation Metrics for Logistic Regression

Evaluating the logistic regression model helps assess its performance and ensure it generalizes well to new, unseen data. The following metrics are commonly used:

1. Accuracy: Accuracy tells us: **How many predictions were correct out of all predictions?**

$$\boldsymbol{Accuracy} = \frac{\text{TruePositives} + \text{True Negatives}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \text{(or)} \frac{\text{Number of correct pridictions}}{\text{Total Number of predicttions}}$$

2. Precision: Precision answers: **Of all students predicted as Pass, how many actually passed?**

$$\textbf{Precision} = \frac{\text{True Positives}}{\text{True positives} + \text{False positives}}$$

3. Recall (Sensitivity or True Positive Rate): Recall answers:

Of all students who actually passed, how many did the model correctly identify?

$$\textbf{Recall} = \frac{\text{True Positives}}{\text{True positives} + \text{False Negatives}}$$

4. Specificity: Specificity answers:

Of all students who actually failed, how many did the model correctly identify as Fail?

$$\textbf{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False positives}}$$

5. F1 Score: F1 Score is the harmonic mean of precision and recall.

$$\textbf{F1 Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

6.2.2 Odds and log-odds

The odds of an event are:

$$Odds = \frac{P}{1-P}$$

Logistic regression assumes that the log-odds are linearly related to the features:

$$log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$$

The coefficient βj represents the change in log-odds for a one-unit increase in Xj, holding other variables constant.

The quantity: e^{β_j} is called the **odds ratio**.

Example

If βj = 0.69, then: $e^{0.69} \cong 2$

A one-unit increase in the feature approximately doubles the odds of the positive class, assuming the other variables remain fixed.

Linear Regression vs Logistic Regression

Feature	Linear Regression	Logistic Regression
Main purpose	Regression	Classification
Output	Continuous numerical value	Probability/class
Example	Predict temperature	Defective/non-defective
Output range	– ∞ to + ∞	0 to 1
Main function	Linear equation	Logistic/sigmoid function
Example output	₹75,000	0.82 probability
Typical loss	Squared-error based	Log-loss / cross-entropy

Problem 1: The following data show the relationship between:

- X1 = Number of hours studied
- X2 = Number of assignments completed
- Y = Final examination score

Student	Study Hours (X1)	Assignments Completed (X2)	Final Score (Y)
S1	1	1	50
S2	2	1	55
S3	3	2	65
S4	4	2	70
S5	5	3	80

Find the multiple linear regression equation and predict the final score of a student who studies for 6 hours and completes 3 assignments.

Solution: **Multiple Linear Regression Equation:**

When there are two predictor variables, the regression equation is: $\widehat{Y} = a + b1\ X1 + b2\ X2$

where:

a = intercept

b1 = coefficient of X1

b2 = coefficient of X2

Our objective is to find: a, b1, b2

Prepare the calculation table: We need the following quantities:

$$X_1,\quad X_2,\quad Y,\quad X_1^2,\quad X_2^2,\quad X_1X_2,\quad X_1Y,\quad X_2Y$$

X_1	X_2	Y	X_1^2	X_2^2	$X_1\ X_2$	X_1Y	X_2Y
1	1	50	1	1	1	50	50
2	1	55	4	1	2	110	55
3	2	65	9	4	6	195	130
4	2	70	16	4	8	280	140
5	3	80	25	9	15	400	240
$\Sigma X_1=15$	**$\Sigma X_2=9$**	**$\Sigma Y=320$**	**$\Sigma X_1^2=55$**	**$\Sigma X_2^2=19$**	**$\Sigma X_1X_2=32$**	**$\Sigma X_1Y=1035$**	**$\Sigma X_2Y=615$**

3. Form the normal equations

For multiple linear regression,

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

The normal equations are:

$$na + b_1\sum X_1 + b_2\sum X_2 = \sum Y$$

$$a\sum X_1 + b_1\sum X_1^2 + b_2\sum X_1X_2 = \sum X_1Y$$

$$a\sum X_2 + b_1\sum X_1X_2 + b_2\sum X_2^2 = \sum X_2Y$$

Since $n = 5$, substitute our values.

4. Solve the three equations

From equation (1):

$$5a + 15b_1 + 9b_2 = 320$$

Divide by 5:

$$a + 3b_1 + 1.8b_2 = 64$$

Therefore,

$$a = 64 - 3b_1 - 1.8b_2$$

Substitute this into equations (2) and (3).

After simplification, we get:

$$10b_1 + 5b_2 = 75$$

or

$$\boxed{2b_1 + b_2 = 15} \tag{4}$$

and

$$5b_1 + 2.8b_2 = 39$$

Now solve these two equations.

First equation

$$5a + 15b_1 + 9b_2 = 320$$

Second equation

$$15a + 55b_1 + 32b_2 = 1035$$

Third equation

$$9a + 32b_1 + 19b_2 = 615$$

Therefore, we have:

$$\boxed{5a + 15b_1 + 9b_2 = 320} \tag{1}$$

$$\boxed{15a + 55b_1 + 32b_2 = 1035} \tag{2}$$

$$\boxed{9a + 32b_1 + 19b_2 = 615} \tag{3}$$

From (4):

$$b_2 = 15 - 2b_1$$

Substitute into the second equation:

$$5b_1 + 2.8(15 - 2b_1) = 39$$

$$5b_1 + 42 - 5.6b_1 = 39$$

$$-0.6b_1 = -3$$

Therefore,

$$\boxed{b_1 = 5}$$

Now:

$$b_2 = 15 - 2(5)$$

$$\boxed{b_2 = 5}$$

Finally:

$$a = 64 - 3(5) - 1.8(5)$$

$$a = 64 - 15 - 9$$

$$\boxed{a = 40}$$

From (4):

$$b_2 = 15 - 2b_1$$

Substitute into the second equation:

$$5b_1 + 2.8(15 - 2b_1) = 39$$

$$5b_1 + 42 - 5.6b_1 = 39$$

$$-0.6b_1 = -3$$

Therefore,

$$\boxed{b_1 = 5}$$

Now:

$$b_2 = 15 - 2(5)$$

$$\boxed{b_2 = 5}$$

Finally:

$$a = 64 - 3(5) - 1.8(5)$$

$$a = 64 - 15 - 9$$

$$\boxed{a = 40}$$

5. Obtain the regression equation

We have:

$$a = 40$$

$$b_1 = 5$$

$$b_2 = 5$$

Therefore, the multiple linear regression equation is:

$$\boxed{\hat{Y} = 40 + 5X_1 + 5X_2}$$

6. Predict the score of a new student

The student:

- Studies for 6 hours
- Completes 3 assignments

Therefore:

$$X_1 = 6$$

$$X_2 = 3$$

Substitute into the regression equation:

$$\hat{Y} = 40 + 5(6) + 5(3)$$

$$= 40 + 30 + 15$$

$$\boxed{\hat{Y} = 85}$$

Therefore, the predicted final examination score is:

$$\boxed{85 \text{ marks}}$$

Practice Problems:

1. Given z = 2 Calculate the probability using the $Sigmoid\ function = \frac{1}{1+e^{-z}}$

2. A logistic regression model produces $z = -1.5$

    1. Calculate the sigmoid probability.
    2. If the classification threshold is 0.5, what is the predicted class?
    3. What happens if the threshold is changed to 0.2?

3. A logistic regression model for predicting whether a student will pass is: $z = -4 + 0.08\,X_1 + 0.5\,X_2$

    where:X1 = study hours, X2= number of practice tests attempted

    For a student who studies **40 hours** and attempts **3 practice tests**:

    1. Calculate z.
    2. Calculate the probability of passing.
    3. Predict Pass/Fail using threshold = 0.5.

4. Given the model: $z = -2 + 0.1\,X$

Calculate the probability of Class 1 for:

1. X=10
2. X=20
3. X=30
4. X=40

At what approximate value of X does the probability become 0.5?

5. A logistic regression model gives these probabilities:

Student	Actual	Predicted Probability
A	1	0.90
B	0	0.80
C	1	0.70
D	0	0.40
E	1	0.30
F	0	0.10

Using a threshold of **0.5**, calculate:

a) TP
b) TN
c) FP
d) FN
e) Accuracy
f) Precision
g) Recall

6. Using the same predictions from Problem 5, change the threshold from **0.5 to 0.7**.

Find:

1. TP
2. TN
3. FP
4. FN
5. Precision
6. Recall

7. A logistic regression model is: $log\left(\frac{P}{1-P}\right) = -3 + 0.5\,X$

For (X = 8):

1. Calculate the log-odds.
2. Calculate the odds.
3. Calculate the probability P.
4. Classify the observation using threshold 0.5.
5. Interpretation of Coefficient.

8. Suppose: $\boldsymbol{log\left(\frac{P}{1-P}\right) = -2 + 0.8\,X}$

1. What does the coefficient **0.8** mean?
2. Calculate the corresponding **odds ratio** $(\mathrm{OR}) = e^{0.8}$
3. Explain what the odds ratio means in practical terms.

Solution:

Given the logistic regression equation:

$$\log\left(\frac{p}{1-p}\right) = -2 + 0.8X$$

where p is the probability of the outcome occurring and X is an independent variable.

1. What does the coefficient 0.8 mean?

The coefficient of X is 0.8.

In logistic regression, the coefficient represents the change in log-odds for a one-unit increase in X, keeping other variables constant.

Therefore, when X increases by one unit, the log-odds increase by 0.8.

$$\boxed{\Delta\text{Log-odds} = 0.8}$$

Interpretation: If X represents study hours, each additional hour of study increases the log-odds of the specified outcome (for example, passing an examination) by 0.8.

2. Calculate the corresponding odds ratio (OR)

The odds ratio is calculated by exponentiating the coefficient:

$$OR = e^{\beta}$$

Here, $\beta = 0.8$. Therefore,

$$OR = e^{0.8}$$

Using a calculator:

$$e^{0.8} \approx 2.2255$$

Hence,

$$\boxed{OR \approx 2.23}$$

3. Explain the odds ratio in practical terms

An odds ratio of 2.23 means that for every one-unit increase in X, the odds of the outcome are multiplied by approximately 2.23, assuming other predictors remain constant.

The percentage increase in odds is:

$$(OR - 1) \times 100\%$$

$$= (2.2255 - 1) \times 100\%$$

$$\boxed{122.55\% \text{ increase in odds}}$$

Practical example: Suppose X is the number of hours a student studies and the outcome is passing an examination. Under this model, each additional hour of study multiplies the odds of passing by approximately 2.23, or increases the odds by about 122.55%.

Important: this does not mean that the probability of passing increases by 122.55%. The change in probability depends on the student's initial probability of passing.

Verify using a numerical example

For $X = 0$:

$$\log\left(\frac{p}{1-p}\right) = -2$$

$$p = \frac{1}{1+e^2} \approx 0.1192$$

The probability is approximately 11.92%.

For $X = 1$:

$$\log\left(\frac{p}{1-p}\right) = -2 + 0.8 = -1.2$$

$$p = \frac{1}{1+e^{1.2}} \approx 0.2315$$

The probability is approximately 23.15%.

X	Log-odds	Probability
0	−2.0	11.92%
1	−1.2	23.15%

The odds have increased by a factor of approximately 2.23, while the probability has increased from 11.92 % to 23.15 %.

Final answers

1. **Coefficient 0.8:** A one-unit increase in X increases the log-odds by 0.8.
2. **Odds ratio:** e^0.8 approximately 2.23.
3. **Practical meaning:** The odds of the outcome increase by approximately 122.55 % per one-unit increase in X, holding other predictors constant.

Problem 9. A medical diagnostic logistic regression model produces:

- TP = 80
- TN = 900
- FP = 50
- FN = 20

Calculate:

1. Accuracy
2. Precision
3. Recall/Sensitivity
4. Specificity
5. F1-score

Solution: Given:

TP=80, TN=900, FP=50, FN=20

First, total observations: N =TP + TN + FP + FN

N = 80 + 900 + 50 + 20 = 1050

1. Accuracy

Accuracy measures the proportion of all predictions that are correct.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Substitute:

$$= \frac{80 + 900}{1050}$$

$$= \frac{980}{1050} = 0.9333$$

Therefore,

$$\boxed{\text{Accuracy} = 93.33\%}$$

Interpretation: The model correctly classifies about 93.33% of all patients.

2. Precision

Precision answers:

> Of all patients predicted as positive, how many actually have the disease?

$$\text{Precision} = \frac{TP}{TP + FP}$$
$$= \frac{80}{80 + 50}$$
$$= \frac{80}{130} = 0.6154$$

Therefore,

$$\boxed{\text{Precision} = 61.54\%}$$

Interpretation: Among patients predicted to have the disease, about **61.54% actually have it.**

3. Recall / Sensitivity

Recall answers:

> Of all patients who actually have the disease, how many did the model correctly identify?

$$\text{Recall} = \text{Sensitivity} = \frac{TP}{TP + FN}$$
$$= \frac{80}{80 + 20}$$
$$= \frac{80}{100} = 0.80$$

Therefore,

$$\boxed{\text{Recall/Sensitivity} = 80\%}$$

Interpretation: The model correctly detects **80% of patients who actually have the disease.**

The remaining 20% are false negatives.

4. Specificity

Specificity answers:

> Of all patients who actually do not have the disease, how many did the model correctly identify as negative?

$$\text{Specificity} = \frac{TN}{TN + FP}$$
$$= \frac{900}{900 + 50}$$
$$= \frac{900}{950}$$
$$= 0.9474$$

Therefore,

$$\boxed{\text{Specificity} = 94.74\%}$$

Interpretation: The model correctly identifies about **94.74% of healthy patients.**

5. F1-Score

F1-score is the harmonic mean of precision and recall:

$$F1 = 2\left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}\right)$$

We have:

$$\text{Precision} = 0.6154$$

$$\text{Recall} = 0.80$$

Therefore,

$$F1 = 2\left(\frac{0.6154 \times 0.80}{0.6154 + 0.80}\right)$$

$$= 2\left(\frac{0.49232}{1.4154}\right)$$

$$\approx 0.6957$$

Therefore,

$$\boxed{F1\text{-score} \approx 69.57\%}$$

Metric	Formula	Value
Accuracy	(TP+TN)/N	93.33%
Precision	TP/(TP+FP)	61.54%
Recall / Sensitivity	TP/(TP+FN)	80.00%
Specificity	TN/(TN+FP)	94.74%
F1-score	2 (Precision× Recall) / (Precision + Recall)	69.57%

1. What is Training Error?

Definition: Training error is the difference between the actual and predicted values for the observations used to train a machine-learning model. It measures how well the model fits the training data.

In simple words: **Training error tells us how well the model fits the data it learned from.**

Suppose we have data from 800 students and use those 800 students to train our model.

After training, we ask the model to predict the scores of the **same 800 students**.

The difference between their actual scores and predicted scores is used to calculate the **training error**.

Training Error = Error on the training data

Simple Example of Training Error

Suppose we want to predict examination marks.

Consider five students:

Student	Actual Marks	Predicted Marks
A	80	78
B	70	72
C	90	85
D	60	63
E	75	74

The model was trained using these students.

For each student, calculate the prediction error:

Error = Actual – Predicted

Student	Actual	Predicted	Error
A	80	78	2
B	70	72	-2
C	90	85	5
D	60	63	-3
E	75	74	1

We can use different measures to calculate the overall training error.

2. What is Testing Error?

Definition: Testing error is the difference between the actual and predicted values for observations that were not used during model training. It measures the model's ability to generalize to unseen data.

In simple words:

Testing error tells us how well the model performs on new, unseen data.

Suppose we have 1,000 students.

We divide them into:

- 800 students → training data
- 200 students → testing data

We train the model using only the 800 training students.

Then we give the model the 200 testing students.

The errors on these 200 students are used to calculate the **testing error**.

Testing Error = **Error on unseen test data**

Student Example

Suppose our model is trained using: 800 students

After training, we evaluate it on: 200 new students

Suppose we obtain: Training MSE = 5, and Testing MSE = 8

This is generally a reasonable situation because the model performs relatively well on both training and unseen data.

Three Important Situations

Situation 1: Underfitting

Suppose: Training Error=20

Testing Error=22

Both errors are high. The model is not learning the underlying pattern sufficiently.

Therefore: **Underfitting**

Typically: **High Bias**

Situation 2: Good Fit

Suppose: Training Error=8

Testing Error=10

Both errors are relatively low and reasonably close.

Therefore: **Good Generalization**

Situation 3: Overfitting

Suppose: Training Error=1

but

Testing Error=25

The model performs extremely well on training data but poorly on unseen data.

Therefore: **Overfitting**

Typically: **High Variance**

Training Error and Testing Error as Model Complexity Increases

Suppose we gradually increase model complexity.

Model Complexity	Training Error	Testing Error
Very simple	High	High
Simple	Decreasing	Decreasing
Moderate	Low	**Lowest**
Complex	Very low	Increasing
Very complex	Almost zero	Very high

Conceptually: Model complexity → Training error decreases

But testing error often behaves differently: Testing error first decreases and then increases

This is the basis of the **underfitting–good fit–overfitting** concept.

Why Does Training Error Usually Decrease?

As we make a model more flexible, it has more ability to fit the training observations.

For example:

Model 1: Straight line: $Y = a + b\,x$

It may not fit complicated data well.

Model 2: Polynomial: $Y = a + b\,x + c\,x^2 + d\,x^3$

It is more flexible.

Model 3: Very high-degree polynomial: $Y = a + b_1\,x + b_2\,x^2 + \cdots + b_{20}\,x^{20}$

It can potentially pass extremely close to every training observation.

Therefore: Training error generally decreases as model flexibility increases

But that doesn't guarantee good performance on unseen data.

Why Can Testing Error Increase?

A highly flexible model can start learning:

- random noise
- outliers
- measurement errors
- accidental patterns
- peculiarities specific to the training sample

These patterns may not exist in future data. Therefore, when new data are presented:

Performance becomes poor

This is overfitting.

Bias-variance trade-off

Prediction error can be decomposed conceptually as:

$$\boldsymbol{E}\left[\left(\boldsymbol{Y}-\hat{\boldsymbol{f}}(\boldsymbol{X})\right)^{\boldsymbol{2}}\right]=(\boldsymbol{Bias})^{\boldsymbol{2}}+\boldsymbol{Variance}+\boldsymbol{irreducible\ error}$$

This decomposition is central to understanding model performance.

Bias:

Bias is the error caused by overly restrictive assumptions.

A high-bias model:

- Is too simple.
- Misses' important patterns.
- Tends to underfit.
- Often has similar errors across different training sets.

Example: fitting a linear model to a highly curved relationship.

Variance

Variance is the sensitivity of a model to the particular training data used.

A high-variance model:

- Is highly flexible.
- Learns small fluctuations in the training data.
- Tends to overfit.
- May produce very different models for different training samples.

Example: an extremely deep decision tree.

Irreducible error

Irreducible error is uncertainty that cannot be removed completely by changing the model.

Sources include:

- Measurement noise.
- Missing variables.
- Human inconsistency.
- Random events.
- Inaccurate labels.

No model can predict a completely random component perfectly.

Trade-off

Increasing model complexity generally:

- Decreases bias.
- Increases variance.

Decreasing model complexity generally:

- Increases bias.
- Decreases variance.

The objective is not to minimize bias or variance individually. The objective is to minimize expected error on new data.

Practical intuition

Imagine aiming at a target:

- High bias, low variance: shots are tightly grouped but far from the centre.
- Low bias, high variance: shots are spread out around the centre.
- Good model: shots are concentrated near the centre.

Model validation

Model validation estimates how well a model is likely to perform on new data.

Training, validation, and test sets

A common arrangement is:

- **Training set:** Used to fit model parameters.
- **Validation set:** Used to choose models, features, thresholds, or hyperparameters.
- **Test set:** Used once for final unbiased evaluation.

Example split:

- 70% training.
- 15% validation.
- 15% test.

The exact proportions depend on dataset size and application requirements.

Important rule

The test set should not be repeatedly used for model decisions. If it is used many times, it gradually becomes part of the model-selection process and no longer provides an unbiased final estimate.

Problems

Problem 1 — Identify Underfitting and Overfitting

Three models are developed to predict the final examination marks of CSE students.

Model	Training MSE	Testing MSE
A	25	27
B	8	10
C	1	35

Questions

1. Which model is underfitting?
2. Which model is overfitting?
3. Which model is the best?
4. Explain the bias and variance characteristics.

Solution

Model A

Training Error = 25

Testing Error = 27

Both errors are high.

The model is unable to learn the important pattern.

Therefore: Model A = Underfitting

It is generally associated with: High Bias + Low Variance

Model B

Training Error = 8

Testing Error = 10

Both errors are relatively low and close to each other.

Therefore: Model B = Good Fit

It provides good generalization.

Model C

Training Error = 1

but

Testing Error = 35

The model performs extremely well on training data but poorly on unseen data.

Therefore: Model C = Overfitting

It is generally associated with: Low Bias + High Variance

Problem 2 — Calculate the Training MSE and Testing MSE

A model predicts examination marks for five training students as follows:

Student	Actual	Predicted
A	80	70
B	60	65
C	90	85
D	70	75
E	50	55

Calculate the training MSE.

For five test students:

Student	Actual	Predicted
F	80	78
G	70	68
H	90	85
I	60	65
J	50	52

Calculate the testing MSE and determine whether the model appears to generalize well.

Solution:

Step 1: Training MSE

Formula: $MSE = \frac{1}{n}\sum_{i=1}^{n}(y - \hat{y})^2$

Errors: 80 - 70 = 10, 60 - 65 = - 5, 90 – 85 = 5, 70 – 75 = - 5, 50 – 55 = - 5

Squared errors: 100, 25, 25, 25, 25

Therefore: $MSE(train) = \frac{100+25+25+25+25}{5} = \frac{200}{5} = 40$

Step 2: Testing MSE

Errors: 80 – 78 = 2, 70 – 68 = 2, 90 – 85 = 5, 60 – 65 = - 5, 50 – 52 = - 2

Squared errors: 4, 4, 25, 25, 4

Therefore: $MSE(test) = \frac{4+4+25+25+4}{5} = \frac{62}{5} = 12.5$

Interpretation

Interestingly, the test error is lower than the training error: 40 > 12.4

This does **not automatically indicate overfitting**.

The particular training sample may simply be more difficult than the test sample.

The important point is that the test error is reasonably low.

Problem 3 — Clear Case of Overfitting

A machine-learning model is used to predict house prices.

The following results are obtained:

Model	Training MSE	Testing MSE
Linear Regression	400	450
Polynomial Regression	100	180
Very High-Degree Polynomial	2	900

Questions

1. Identify underfitting/overfitting.
2. Which model should be selected?
3. Explain using bias and variance.

Solution:

Linear Regression

Training MSE= 400, Testing MSE = 450

Both errors are relatively high.

This suggests that the model is too simple, and High Bias → Underfitting

Polynomial Regression

Training MSE = 100, Testing MSE = 180

Both are substantially lower. This is the best model among the three.

Good bias - variance balance

Very High-Degree Polynomial

Training MSE = 2

but

Testing MSE = 900

The model almost perfectly fits training data but fails on new data.

Therefore: Overfitting

This corresponds to: Low Bias + Very High Variance

Best model is Polynomial Regression

because its **testing error is lowest**.

Problem 4 — Calculate Bias Squared and variance

Suppose the true value of Y at a particular input X is: f (X) = 80

Five different training samples are used to train the same model.

The predictions at X are: 70, 74, 78, 76, 72

Calculate:

1. Mean prediction
2. Bias
3. Bias squared
4. Variance

Solution:

Step 1: Mean prediction

$$E\left[\hat{f}(x)\right] = \frac{70+74+78+76+72}{5} = \frac{370}{5} = 74 = Mean$$

Step 2: Bias

$$Bias = E\left[\hat{f}(x) - f(x)\right] = 74 - 80 = -6$$

The model systematically underpredicts by 6 marks.

Step 3: Bias squared

$$\boldsymbol{Bias^2 = (-6)^2 = 36}$$

Interpretation

The model has a systematic error of 6 marks.

The contribution of squared bias is: 36

Mean = 74

Calculate the variance of the predictions.

$$Variance = \frac{1}{n}\sum\left[\hat{f}(x) - E(\hat{f}(x)\right]^2$$

The deviations from the mean are:

Prediction	Mean	Deviation	Squared deviation
70	74	-4	16
74	74	0	0
78	74	4	16
76	74	2	4
72	74	-2	4

Therefore:

$$Variance = \frac{16 + 0 + 16 + 4 + 4}{5} = 8$$

Interpretation

The predictions are reasonably close to one another.

Therefore, the model has relatively low variance.

Problem 5— Complete Bias–Variance Calculation

Suppose the true value is: f (X) = 80

A model produces the following predictions from five different training datasets: 70, 74, 78, 76, 72

Calculate:

1. Irreducible error if $\sigma^2 = 10$
2. Total expected prediction error

Solution: From Problems 4:

$E\left[\hat{f}(x)\right] = 74 = Mean$

$Bias = E\left[\hat{f}(x) - f(x)\right] = -6$

$\boldsymbol{Bias^2 = (-6)^2 = 36}$

Variance = 8

Irreducible error: $\sigma^2 = 10$

The bias-variance decomposition is: $Expected\ Test\ Error = Bias^2 + \text{Variance} + (\text{Irreducible error})\ \sigma^2$

Therefore: Expected Test Error = 36 + 8 + 10 = 54

Final answer

Component	Value
Mean prediction	74
Bias	-6
Bias²	36

Component	Value
Variance	8
Irreducible error	10
Expected test error	**54**

Problem 6 — Compare Two Models Using Bias and Variance

Two models are being considered for predicting student examination scores.

Model A: Bias = 8, Variance = 4, Irreducible Error = 9

Model B: Bias = 2, Variance = 25 Irreducible Error = 9

Calculate the expected test error for each and determine which model is preferable.

Solution:

Model A, First calculate bias squared: Bias^2 = 8^2 = 64

Therefore: Test Error = 64 + 4 + 9 = 77

Model B, Bias^2 = 2^2 = 4

Therefore: Test Error = 4 + 25 + 9 = 38

Although Model B has much higher variance, its bias is substantially lower.

Therefore: 38 < 77

So: Model B is preferable

Important lesson: A model with slightly higher variance can sometimes have a **lower overall test error** because its bias is much smaller.

That is the essence of the: Bias – Variance Trade – off

Problem 7 — Identify the Best Model from Bias and Variance

Three models have the following characteristics:

Model	Bias	Variance	Irreducible Error
A	10	2	5
B	5	10	5
C	2	30	5

Calculate expected test error and identify the best model.

Solution: The formula is: EPE = $Bias^2$ + Variance + Irreducible Error

Model A

$Bias^2 = 10^2 = 100$, EPE = 100 + 2 + 5 = 107

Model B

$Bias^2 = 5^2 = 25$, EPE = 25 + 10 + 5 = 40

Model C

$Bias^2 = 2^2 = 4$, EPE = 4 + 30 + 5 = 39

Therefore: Model C is slightly better

because: 39 < 40 < 107

Notice that Model C has the highest variance, but it is very low bias compensates for it.

Problem 8: Find the Optimum Model Complexity

Consider the following models:

Model Complexity	Training Error	Testing Error
1	40	45
2	25	30
3	15	20
4	10	**12**
5	6	15
6	3	25
7	1	40

Questions

1. Which model is underfitting?
2. Which model is overfitting?
3. Which model has the best generalization?
4. At what complexity should the model be selected?

Solution:

The testing errors are: 45, 30, 20, 12, 15, 25, 40

The minimum testing error is: 12 at complexity 4.

Therefore: Best model = Model 4

Underfitting

Models 1 and possibly 2 have relatively high training and testing errors.

For example, Model 1: Training = 40, Testing = 45

indicates: Underfitting

Overfitting

At high complexity:

Training Error → 1

but

Testing Error → 40

Therefore Models 6 and 7 clearly demonstrate overfitting.

Problem 9 — Calculate the Generalization Gap

The **generalization gap** is a useful practical measure:

Generalization Gap = Testing Error - Training Error

Suppose four models have:

Model	Training Error	Testing Error
A	20	25
B	10	13
C	3	15
D	1	30

Calculate the generalization gap for each model.

Solution:

Model A

25 – 20 = 5, Gap = 5

Model B

13 – 10 = 3, Gap = 3

Model C

15 – 3 = 12, Gap = 12

Model D

30 – 1 = 29, Gap = 29

Interpretation

Model	Gap	Interpretation
A	5	Moderate
B	**3**	**Best generalization**
C	12	Possible overfitting
D	**29**	**Severe overfitting**

A large gap between training and testing error is a strong warning sign for overfitting.

Problem 11 — Classification Example

A model is used to classify whether a student will **pass (1) or fail (0)**.

The following results are obtained:

Model	Training Accuracy	Testing Accuracy
A	65%	63%
B	90%	87%
C	99%	70%

Identify the models.

Solution:

Model A

Training = 65%, Testing = 63%

Both are low.

Underfitting

Likely: High Bias

Model B

Training = 90%, Testing = 87%

Both are high and close.

Good Model

Model C

Training = 99%

but: Testing = 70%

There is a large performance gap.

Overfitting

Likely: High Variance

Problem 12 — A Complete Real-Life Problem

A company develops three machine-learning models to predict whether a customer will **cancel a subscription**.

The results are:

Model	Training Accuracy	Testing Accuracy	Training Error	Testing Error
A	70%	68%	30%	32%
B	94%	91%	6%	9%
C	99%	72%	1%	28%

Questions

1. Identify underfitting.
2. Identify overfitting.
3. Identify the best model.
4. Which model has high bias?
5. Which model has high variance?
6. Explain the trade-off.

Solution:

Model A

Training accuracy: 70%

Testing accuracy: 68%

Both are relatively low.

Therefore: Model A = Underfitting

High bias.

Model B

Training accuracy: 94%

Testing accuracy: 91%

The model performs well on both training and unseen data.

Therefore: Model B = Good model

It has a good bias-variance balance.

Model C

Training accuracy: 99%

Testing accuracy: 72%

The model has nearly memorized the training data.

Therefore: Model C = Overfitting

High variance.

Cross-validation:

Cross-validation is a repeated training-and-validation procedure used to estimate model performance and select hyperparameters.

In K - fold cross-validation:

1. Divide the data into K approximately equal folds.
2. Use one-fold as the validation set.
3. Train the model on the other K−1 folds.
4. Compute the validation error.
5. Repeat until every fold has served as the validation set.
6. Average the K validation errors.

The estimated cross-validation error is:

$$\text{CV Error} = \frac{1}{K}\sum_{k=1}^{K}\text{Error}_k$$

This procedure allows every observation to be used for validation once and for training multiple times. Common choices include 5-fold and 10-fold cross-validation.

Example

For 5-fold cross-validation:

- Fold 1 validates while folds 2–5 train.
- Fold 2 validates while folds 1, 3–5 train.
- Continue through fold 5.
- Average the five validation errors.

Choosing K

Choice	Advantages	Limitations
Small K, such as 2 or 5	Faster computation	May give a more biased estimate
K=10	Often a practical balance	More computation
Leave-one-out CV	Uses almost all data for each training run	Computationally expensive and can have high variance

The choice depends on dataset size, computational resources, and the stability of the model. The usual goal is not to choose the theoretically largest K, but to obtain a reliable estimate at reasonable cost.

Stratified cross-validation

For classification, especially with imbalanced classes, each fold should preserve approximately the same class proportions.

For example, if only 5% of transactions are fraudulent, every fold should contain a similar percentage of fraudulent transactions. This is called **stratified K-fold cross-validation**.

Time-series validation

Randomly mixing observations is inappropriate for many time-dependent problems.

For stock prices, sensor readings, demand forecasting, or weather prediction:

- Train on earlier observations.
- Validate on later observations.
- Do not use future information to predict the past.

A rolling or expanding-window validation scheme is more appropriate.

Data leakage

Data leakage occurs when information unavailable at prediction time enters the training process.

Examples:

- Using a patient's discharge information to predict admission risk.

- Normalizing the complete dataset before splitting into training and test sets.
- Using future sales data to predict current demand.
- Selecting features using the entire dataset before cross-validation.

Correct procedure

For each cross-validation training fold:

1. Fit preprocessing steps only on the training part.
2. Apply the fitted preprocessing to the validation part.
3. Train the model on the transformed training data.
4. Evaluate on the transformed validation data.

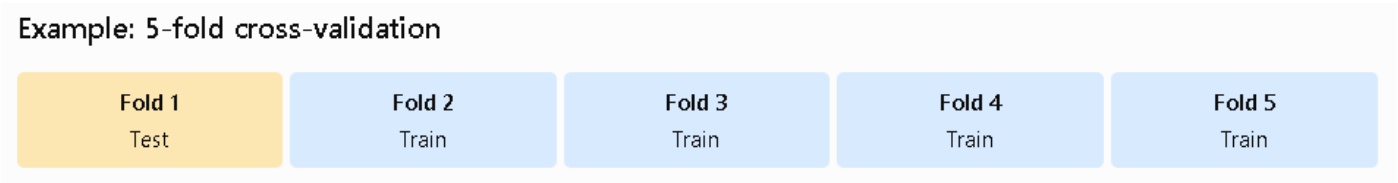
In practice, preprocessing and modelling should be placed in a single pipeline.

K-Fold Cross-Validation for Classification: Problems with Solutions

K-fold cross-validation is an important technique in statistical learning used to evaluate classification models, such as logistic regression, decision trees, K-nearest neighbours (KNN), and support vector machines (SVM).

1. Basic concept

In K - fold cross-validation, the dataset is divided into K approximately equal folds. The model is trained K times. Each time, one-fold is used for testing and the remaining (K – 1) fold are used for training.



In the next round, Fold 2 becomes the test fold; this continues until every fold has been used for testing once.

The average classification accuracy is $\mathbf{Mean\ Accuracy} = \frac{1}{K}\sum_{i=1}^{K} A_i$

where A_i is the accuracy obtained on fold i.

Problem 1: Calculate the average classification accuracy

A computer science student builds a logistic regression model to predict whether an email is spam. The dataset is evaluated using 5-fold cross-validation. The classification accuracies are:

Fold	Accuracy
1	80%
2	85%
3	90%
4	75%
5	80%

Find the mean accuracy.

Solution:

Using the formula, $\bar{A} = \frac{A_1+A_2+A_3+A_4+A_5}{5}$

$$\bar{A} = \frac{80+85+90+75+80}{5} = \frac{410}{5} = 82\ \%$$

Answer: The estimated mean classification accuracy is 82%.

Interpretation: across the five test folds, the model correctly classified an average of 82% of the emails.

Problem 2: Calculate mean accuracy and mean error rate

A model predicts whether students will pass or fail an examination. A 4-fold cross-validation procedure produces the following results.

Fold	Correct predictions	Total test cases
1	18	20
2	16	20
3	17	20
4	19	20

Calculate the accuracy of each fold, mean accuracy, and mean error rate.

Solution:

Fold 1:

$$A_1 = \frac{18}{20} \times 100 = 90\ \%$$

Fold 2:

$$A_2 = \frac{16}{20} \times 100 = 80\ \%$$

Fold 3:

$$A_3 = \frac{17}{20} \times 100 = 85\ \%$$

Fold 4:

$$A_4 = \frac{19}{20} \times 100 = 95\ \%$$

$$\textbf{\textit{Mean Accuracy}} = \bar{A} = \frac{90 + 80 + 85 + 95}{4} = \frac{350}{4} = 87.5\ \%$$

Mean Error Rate:

Error rate =100% - 87.5 % = 12.5 %

Answer: Mean accuracy = 87.5%; mean error rate = 12.5%.

Problem 3: Calculate mean precision, recall and F1-score

A fraud-detection classification model is evaluated using 3-fold cross-validation. The results are:

Fold	Precision	Recall
1	80%	70%
2	75%	80%
3	85%	75%

Calculate the mean precision, mean recall, and mean F1-score. For each fold, calculate F1 first and then average the three F1-scores.

Solution:

$$\boldsymbol{Mean\ Precision} = \frac{\mathbf{80+75+85}}{\mathbf{3}} = \mathbf{80}\%$$

$$\boldsymbol{Mean\ Recall} = \frac{\mathbf{70+80+75}}{\mathbf{3}} = \mathbf{75}\%$$

$$\boldsymbol{F1\ Score\ formula} = \frac{\mathbf{2\ P\ R}}{\boldsymbol{P+R}} = \mathbf{80}\%$$

Here, P and R, are expressed as percentages.

Fold 1:

$$F1_1 = \frac{2(80)(70)}{80+70} = \frac{11200}{150} = 74.67\ \%$$

Fold 2:

$$F1_2 = \frac{2(75)(80)}{75+80} = \frac{12000}{155} = 77.42\ \%$$

Fold 3:

$$F1_3 = \frac{2(85)(75)}{85+75} = \frac{12750}{150} = 79.69\ \%$$

Mean F1 - Score

$$\overline{F1} = \frac{74.67+77.42+79.69}{2} = 77.26\ \%$$

Answer: Mean precision = 80%, mean recall = 75%, and mean fold-wise F1-score ≈ 77.26%.

Note: averaging the fold-wise F1-scores is not necessarily the same as calculating F1 from the mean precision and mean recall.

Problem 4: Select the best classification model using 5-fold cross-validation

A university compares three classification algorithms for predicting student performance.

Fold	Logistic regression	Decision tree	KNN
1	82%	90%	78%
2	84%	65%	80%
3	83%	92%	82%
4	81%	60%	79%

5	85%	93%	81%

Find the average accuracy of each model and recommend one.

Solution:

$$\text{Logistic regression: } A_L = \frac{82+84+83+81+85}{5} = 83\ \%$$

$$\text{Decision tree: } A_D = \frac{90+65+92+60+93}{5} = 80\ \%$$

$$\text{KNN: } A_K = \frac{78+80+82+79+81}{5} = 80\ \%$$

Conclusion: Logistic regression is the preferred model based on mean accuracy. The decision tree has highly variable results across folds, so its stability deserves attention. A final choice should also consider precision, recall, and other relevant metrics.

Problem 5: Stratified K-fold cross-validation

A dataset contains 100 student records:

- 80 students passed (Class 1).
- 20 students failed (Class 0).

The dataset is divided using 5-fold stratified cross-validation.

Questions:

1. How many records are in each fold?
2. Approximately how many passed and failed students should each fold contain?
3. Why is stratification useful?

Solution

Total records = 100 and K = 5.

Records per fold:

$$\frac{100}{5} = 20$$

Each fold should approximately preserve the original class proportions.

Passed students per fold:

$$\frac{80}{5} = 16$$

Failed students per fold:

$$\frac{20}{5} = 4$$

Thus, each fold should contain approximately 16 passed and 4 failed students.

Why stratification matters: It maintains a similar class distribution in every fold. This is especially helpful for imbalanced datasets, such as fraud detection, disease screening, and rare-event classification.

Problem 6: Calculate the overall accuracy from pooled predictions

Suppose a 3-fold classification experiment produces these confusion-matrix counts on the respective test folds.

Fold	TP	TN	FP	FN
1	30	50	10	10
2	35	45	5	15
3	25	55	10	10

Calculate the overall accuracy by combining all test-fold predictions.

Solution:

First, add the correct predictions:

$$TP_{Total} = 30 + 35 + 25 = 90$$

$$TN_{Total} = 50 + 45 + 55 = 150$$

Total correct predictions: 90 + 150 = 240

Total observations: N = 3 (30 + 50 + 10 + 10) = 300

Therefore, Accuracy = (TP + TN) / N

$$Accuracy = \frac{\text{Total correct predictions}}{\text{Total observations}} = \frac{240}{300} \times 100 = 80\ \%$$

Answer: The pooled cross-validation accuracy is 80%.

This calculation weights each observation equally. When fold sizes differ, pooled accuracy may differ from the unweighted mean of fold accuracies.

Real-life applications

1. Healthcare

Applications

- Predicting disease risk.
- Classifying medical images.
- Estimating hospital readmission probability.
- Forecasting patient length of stay.
- Detecting abnormal heart rhythms.
- Predicting treatment response.

Regression example

Predict the number of days a patient may remain in a hospital using:

- Age.
- Diagnosis.
- Vital signs.

- Lab measurements.
- Previous medical history.

Logistic-regression example

Predict whether a patient is at high risk of a condition: P (disease = 1 | X)

Important considerations

- False negatives may be more dangerous than false positives.
- Patient privacy must be protected.
- Data may contain demographic bias.
- A model should support, not blindly replace, clinical judgment.
- Predictions should be explainable and validated on independent populations.

2. Banking and finance

Applications

- Credit-risk prediction.
- Fraud detection.
- Loan default prediction.
- Customer churn analysis.
- Cash-demand forecasting.
- Portfolio-risk estimation.

Example

A logistic-regression model may estimate:

$$P(\text{default} = 1 \mid \text{income}, \text{credit history}, \text{debt}, \text{age})$$

A bank may choose a threshold based on the relative costs of approving a risky loan and rejecting a reliable customer.

Engineering challenges

- Fraud patterns change over time.
- Fraud cases are often rare.
- The model must be resistant to manipulation.
- Evaluation should include precision, recall, cost, and fairness—not accuracy alone.

3. E-commerce and recommendation systems

Applications

- Predicting customer purchase probability.
- Recommending products.
- Estimating demand.

- Predicting customer lifetime value.
- Identifying users likely to leave.
- Personalizing search results.

Example

A regression model may predict expected revenue from a customer:

Y^ = f (past purchases, visits, product views, response to offers)

A classification model may predict whether the customer will purchase within the next seven days.

Practical issue

Historical recommendations influence what users see and buy. Therefore, the data may contain feedback loops: products shown more often receive more clicks, which makes them appear even more popular.

4. Transportation and logistics

Applications

- Travel-time prediction.
- Delivery-time estimation.
- Route selection.
- Ride-demand forecasting.
- Predictive maintenance.
- Traffic congestion prediction.

Regression example

Predict delivery time using:

- Distance.
- Traffic level.
- Weather.
- Number of stops.
- Vehicle type.
- Time of day.

A model should be evaluated separately for urban and rural routes because overall performance can hide poor performance in one group.

5. Cybersecurity

Applications

- Spam detection.
- Malware classification.
- Intrusion detection.

- Phishing detection.
- User-behaviour analysis.
- Anomaly detection.

Logistic-regression example

Predict whether a network connection is malicious using:

- Packet count.
- Connection duration.
- Port number.
- Data volume.
- Protocol.
- Historical behaviour.

Important issue

Cybersecurity is adversarial. Attackers may deliberately change behaviour to fool the model. Models therefore require continuous monitoring and periodic retraining.

6. Manufacturing and IoT

Applications

- Predictive maintenance.
- Defect detection.
- Quality control.
- Energy optimization.
- Equipment failure prediction.
- Process monitoring.

Example

A regression model predicts machine temperature or energy consumption. A classification model predicts whether a machine will fail during the next operating period.

Sensor data often contains:

- Missing values.
- Noise.
- Time dependence.
- Sensor drift.
- Changing operating conditions.

A model should therefore be validated under realistic production conditions rather than only on randomly selected historical records.

7. Education technology

Applications

- Predicting student performance.
- Early identification of students needing support.
- Recommendation of learning materials.
- Estimating course completion probability.
- Detecting unusually difficult topics.

Example

A logistic-regression model can estimate the probability that a student will complete a course using:

- Login frequency.
- Assignment submission history.
- Quiz attempts.
- Forum participation.
- Time spent on lessons.

Ethical considerations

Student models should not permanently label students. Predictions should trigger supportive interventions, not automatic punishment.

8. Natural-language processing

Applications

- Sentiment classification.
- Email filtering.
- Text categorization.
- Search ranking.
- Intent detection in chatbots.
- Toxicity detection.

A logistic-regression classifier can use word counts, TF-IDF features, or embeddings to classify documents.

Example:

P (positive sentiment = 1| text features)

The model may perform differently across languages, dialects, and writing styles. Evaluation should therefore include representative test data.

9. Computer vision

Applications

- Face detection.

- Object recognition.
- Quality inspection.
- Medical-image analysis.
- Traffic-sign recognition.
- Satellite-image classification.

A simple statistical-learning pipeline may include:

1. Image acquisition.
2. Image preprocessing.
3. Feature extraction.
4. Model training.
5. Validation.
6. Error analysis.
7. Deployment monitoring.

Although modern computer vision frequently uses deep learning, the ideas of training error, generalization, overfitting, validation, and bias-variance remain fundamental.

A complete example for students

Problem: A company wants to predict whether an online customer will cancel a subscription.

Features

- Customer age.
- Monthly subscription cost.
- Number of support requests.
- Login frequency.
- Contract duration.
- Payment history.

Target: $Y = \begin{cases} 1, \text{customer cancels} \\ 0, \text{customer remains} \end{cases}$

Step 1: Explore the data

Students should inspect:

- Missing values.
- Class proportions.
- Outliers.
- Duplicate records.
- Feature distributions.

- Relationships among variables.

Step 2: Split the data

Create training and final test sets. Keep the test set untouched until final evaluation.

Step 3: Build a baseline

A baseline may predict the majority class for every customer. The machine-learning model should perform better than this baseline.

Step 4: Train logistic regression

Fit: $P(Y = 1 \mid X) == \frac{1}{1+e-(\beta 0+\beta 1X1+\cdots+\beta pXp)}$

Step 5: Perform cross-validation

Use stratified 5-fold cross-validation to compare:

- Logistic regression without regularization.
- Logistic regression with ridge regularization.
- Logistic regression with lasso regularization.
- Decision tree.
- Random forest.

Step 6: Select an evaluation metric

If the company wants to identify as many likely cancellations as possible, recall may be more important than accuracy.

If retention campaigns are expensive, precision may be more important.

Step 7: Select the threshold

A threshold of 0.5 is not automatically optimal. The business can choose another threshold based on campaign cost and expected benefit.

Step 8: Evaluate once on the test set

Report:

- Confusion matrix.
- Precision.
- Recall.
- F1-score.

Step 9: Monitor after deployment

Track:

- Changes in customer behaviour.
- Changes in cancellation rates.
- Model performance over time.

- Fairness across groups.
- Data drift.
- Incorrect predictions.
- The evaluation metric.
- The cost of false positives and false negatives.
- Possible sources of bias.
- How the model will be validated.